Amanda Siriwardane

Department of Computer Engineering, University of Peradeniya

amandasiriwardane01@gmail.com | +94 77 364 9650 | Colombo, Sri Lanka
linkedin.com/in/amanda-siriwardane | amandasiriwardane.github.io



Profile

Creative-minded computer engineering undergraduate at the University of Peradeniya, who recently completed a Software Engineering internship on backend systems for a security-standards product. Proactive leader with experience managing student societies, proven ability to quickly learn and adapt to new environments, seeking a permanent role to contribute strong fundamentals, collaborate with experienced engineers, and grow in modern product development environments.

Experience

Software Engineering Internship, *IAM Team at WSO2 LLC* April 2026 – Oct 2026
- Onboarding Project - Paws & Homes-pet adoption platform using WSO2 products.
- Automated Pre-migration assessment process of WSO2 Identity Server
- Implemented SSF & CAEP capability for WSO2 Identity Server event sharing

Undergraduate Teaching Assistant, *Department of Computer Engineering, UoP*
- Programming for Engineers (CO1810) - Assisted 100+ students with Python and C fundamentals, supported lab sessions, and provided code reviews — April 2025 – Aug 2025
- Data Structures and Algorithms II (CO2030) - Java — Nov 2025 – March 2026

Projects

Implementing SSF/CAEP Event Delivery for WSO2 Identity Server | Internship Project Aug 2026 – Oct 2026
Java, OSGi, REST API Design, OAuth2, JWS/JWT, WSO2 Identity Server
- Designed and built a new connector implementing the SSF Stream Management REST API, enabling WSO2 Identity Server to deliver CAEP security events to external receivers over an OAuth2-protected, standards-based transport.
- Architected the service layer to delegate CRUD operations to the existing WebhookManagementService, avoiding duplicated persistence logic.
- Redesigned Identity Server's event-publisher resolution from a single, non-deterministic lookup to a per-event-profile map, enabling the new JWS-signing SSF publisher to run alongside the existing HTTP publisher with zero regression.

Automating Pre-Migration Assessments for WSO2 Identity Server | Internship Project May 2026 – July 2026
SQL, Docker, MySQL, WSO2 Identity Server, CRE AI Agent (Claude + MCP)
- Built an automated system replacing manual calls with diagnostic SQL scripts to surface feature usage before migration.
- Authored and live-verified SQL query sets across major IS upgrades (5.10.0 to 7.0.0) against Docker-hosted IS instances.
- Developed an internal Claude Agent Skill that dynamically generates a dialect-aware SQL script per version pair with traceability and risk flags.

Paws & Homes: A Pet Adoption Portal | Internship Onboarding Project | *WSO2* May 2026 (2 Weeks)
React, Node.js, Express, PostgreSQL, WSO2 Asgardeo, WSO2 Choreo
- Built a full-stack SaaS pet adoption platform to gain hands-on experience with modern cloud-native architecture.
- Implemented OIDC-based authentication and JWT validation with WSO2 Asgardeo, supporting 3 distinct user roles.
- Developed a RESTful API backend with a PostgreSQL database, securing endpoints with role-based access control.
- Deployed and managed the application using WSO2 Choreo for CI/CD and API management.

Multimodal Phenotyping in Knee Osteoarthritis Using VLMs | Research | *Group* Sep 2025 – March 2026
Python, PyTorch, BioMedCLIP, HDBSCAN, Explainable AI (Grad-CAM)
- Worked on an unsupervised multimodal framework to identify novel knee osteoarthritis phenotypes by combining radiographic images and clinical text data by utilizing Vision-Language Models (BioMedCLIP) to align medical imaging features with corresponding clinical descriptions.
- Implemented HDBSCAN clustering to identify latent disease subgroups beyond standard severity scales.
- Validated phenotypes using XAI techniques (Grad-CAM, textual explanations) for clinical transparency.

Safe Plus (Smart Safety Helmet) | 3rd Year Project | *Group* Nov 2024 – July 2025
ESP32, AWS IoT Core, MERN Stack, JWT Authentication, 2FA (Speakeasy), Netlify, Railway
- Built a smart safety helmet detecting industrial hazards with real-time alerts via ESP32 → AWS IoT → Backend → Dashboard/Mobile.

- **Contribution:** *Led dashboard development*- Implemented secure authentication (JWT, bcrypt, 2FA), worker management, and real-time sensor visualization. *Managed full deployment*- Frontend (Netlify), Backend (Railway), and domain configuration.
- **Achievements:** Showcased at EngEx 2025; Finalist in IEEE Sri Lanka FYP Arena 2025.

MediGo - Pharmacy Delivery Platform | Individual Learning Project | url: Aug 2025 – Dec 2025
MERN Stack, Redux Toolkit, Material-UI, Cloudinary, Multer, RESTful APIs
- Developed a multi-role pharmacy delivery platform serving 3 user types with 15+ RESTful API endpoints and role-based auth, Cloudinary-powered product management and centralized state management using Redux Toolkit.
- Designed normalized MongoDB schemas with relational models and deployed to production using Vercel (Frontend) and Render (Backend), configuring custom SPA routing rules.

Production Monitoring Dashboard | 2nd Year Project | *Group* April 2024 – Oct 2024
JavaScript, React, Python, MongoDB, Real-time Data Processing
- Built a real-time monitoring system for 24 plastic toy machines, providing live production updates and emergency alerts to minimize downtime. Implemented live data visualization and alert systems with scalable backend architecture & maintained historical production records for analytics and performance tracking.

Education

University of Peradeniya, BScEngHons specializing in Computer Engineering Feb 2022 – Oct 2026
- GPA: 3.56/4.0
- **Coursework:** Data Structures and Algorithms, Object Oriented Programming, Machine Learning and Data Mining , Web Development, Database Systems, Neural Networks

President Girls' School, Colombo, Secondary Education | Physical Stream | **Z- Score:** 1.9970 | **Island Rank:** 1000
Achievement: First student in school history to qualify for Engineering admission at a State University.

Technical Skills

- **Programming Languages:** JavaScript, Java, Python, C, SQL
- **Frontend/Backend Development:** React.js, Redux Toolkit, HTML, CSS, Node.js, Express.js, RESTful API Design, OSGi
- **Databases & Cloud :** MongoDB, MySQL, AWS, Vercel, Netlify, Render, Docker
- **Identity & Access Management:** OAuth2, OIDC, JWT/JWS, WSO2 Identity Server, WSO2 Asgardeo, Shared Signals Framework (SSF) / CAEP
- **AI & Data:** Claude/MCP agent tooling, PyTorch, Vision-Language Models, OpenCV, NumPy
- **Tools & Platforms:** Git, GitHub, Postman

Leadership and Extra-Curricular Activities

IEEE - Computer Society Student Chapter: Chairperson *(Nov 2025 - Present)*, Design Team Lead *(Aug 2024 - Nov 2025)*
AIESEC in University of Peradeniya: Brand Manager & Marketing Specialist - Outgoing Global Volunteering *(Feb 2023 - Jan 2024)*, Team Member - H4TF *(Aug 2022 - Feb 2023)*
13th Shanghai International Youth Interactive Friendship Camp, China: Selected as one of 5 representatives from Sri Lanka to engage in cross-cultural collaboration with youth from 25+ countries. *(July 2017)*
Team Player(Badminton) - University of Peradeniya, President Girls' School
Member - Rotaract Club of University of Peradeniya

Achievements & Certifications

Super Contributor Badge : Hacktoberfest 2025 - Open Source Contribution Oct 2025
Pre-Coders : Finalist in a 6-hour coding competition with 80+ teams Aug 2024
Coders v11.0: 12-hour all-island coding competition with 300+ teams Oct 2024
Half Color Award : for Badminton, UoP Jan 2024
You Tube Labs - AWS Basics | KodeKloud 2026
AWS Technical Essentials | AWS Skill Builder 2025
Artificial Intelligence For beginners | Alison 2024

Referees

Dehami Koswatte
Associate Technical Lead
WSO2 (Pvt) Ltd
Email: dehami@wso2.com

Dr. Isuru Nawinne
Department of Computer Engineering
Faculty of Engineering, University of Peradeniya
Email: isurunawinne@eng.pdn.ac.lk